

MESNA POS SYSTEM

100% Offline-First Multi-Register Point of Sale

Official Installation & User Setup Guide

1. System Overview

Mesna POS is a high-performance retail system built specifically to operate in environments with unreliable or zero internet connectivity. By using a decentralized peer-to-peer (P2P) architecture, multiple checkout registers sync inventory and transaction records in real-time over the local network without requiring any cloud database.

Key Features & Capabilities:

- **P2P Mesh Sync:** Automatic network discovery using UDP local broadcasting (port 41234).
- **Local AI Voice POS:** Speaks and understands Urdu, Hindi, and English command phrases offline using Whisper-Tiny AI.
- **100% Offline-First:** Zero cloud database dependency, running a local high-performance SQLite engine managed via Prisma ORM.
- **Rich Admin Panel:** Includes real-time sales charts, admin panels, inventory management, and receipt printing templates.

2. Hardware & Software Requirements

- **Operating System:** Windows 10 / 11 (recommended) or macOS / Linux.
- **Required Runtime:** Node.js (v18.0.0 or higher) - Core runtime engine.
- **Web Browser:** Google Chrome, Microsoft Edge, or any modern web browser.
- **Receipt Printing:** Any standard POS Thermal Receipt Printer (optional, prints receipts).
- **Voice Input:** A standard USB or built-in Microphone (optional, for Voice POS commands).

3. Option A: Quick Start (Recommended)

The project includes an automated Windows launcher script that installs and boots up all necessary components automatically with one click. This is the easiest way to launch the system.

- **Step 1:** Double-click the file named `start-pos.bat` located in the main project folder.
- **Step 2:** The script will launch two command prompt windows in the background (one for the backend database server, and one for the frontend React user interface).
- **Step 3:** The launcher will automatically open your default browser to `http://localhost:5173` once initialization is complete.

Do Not Close Command Prompts!

Keep the two command prompt windows open while using the POS. Closing them will shut down the database and API servers. To stop the application, simply close the windows or press `Ctrl+C` in them.

4. Option B: Manual Setup & Launch

If you are a developer, on macOS/Linux, or prefer launching from a terminal, follow these steps:

Step 1: Set Up Backend Services

Open your terminal, navigate to the backend directory, install packages, and initialize SQLite:

```
cd backend
npm install
npx prisma migrate dev --name init
npm run seed      # (Optional: seeds sample products & admin PIN 1234)
npm run start
```

Step 2: Set Up Frontend Interface

Open a second terminal window or tab, navigate to the frontend folder, and launch the Vite client:

```
cd frontend
npm install
npm run dev
```

Now, open `http://localhost:5173` in your web browser to access the POS dashboard.

5. Connecting Multiple Registers (LAN Sync)

Mesna POS can run on multiple checkout terminals simultaneously. They discover each other and synchronize transactions and inventory levels automatically without an internet connection.

- **Step 1:** Ensure all computer registers are connected to the exact same Wi-Fi router or local LAN switch.
- **Step 2:** Launch Mesna POS on each laptop/computer using start-pos.bat or the manual setup commands.
- **Step 3:** The machines will automatically discover each other via UDP broadcasting on port 41234.
- **Step 4:** When a transaction is made on Register A, Register B will receive it instantly and deduct its local stock automatically.

Firewall Requirements

Ensure your Windows Defender Firewall (or third-party antivirus firewall) allows traffic on UDP port 41234 and TCP ports 3001 and 5173. If registers cannot find each other, double-check your router's AP isolation setting.

6. Local Voice POS Quick Reference

To use voice commands, click the floating green microphone icon in the bottom right corner of the POS screen, speak naturally in English, Urdu, or Hindi, and click the microphone again to process.

- **Add Product:** Say 'Add Coca Cola' or 'Doodh add karo'.
- **Apply Discount:** Say 'Discount 10 percent' or 'Das percent discount'.
- **Remove Item:** Say 'Remove milk' or 'Doodh delete karo'.
- **Checkout & Pay:** Say 'Checkout', 'Confirm payment card' or 'Bill banao'.
- **Clear Cart:** Say 'Clear cart' or 'Khali karo'.

Need Support or Custom Development?

Check README.md in the root directory or consult your system administrator.